

Nanorings neural SDE

Analysis

Date revised
October 29, 2025

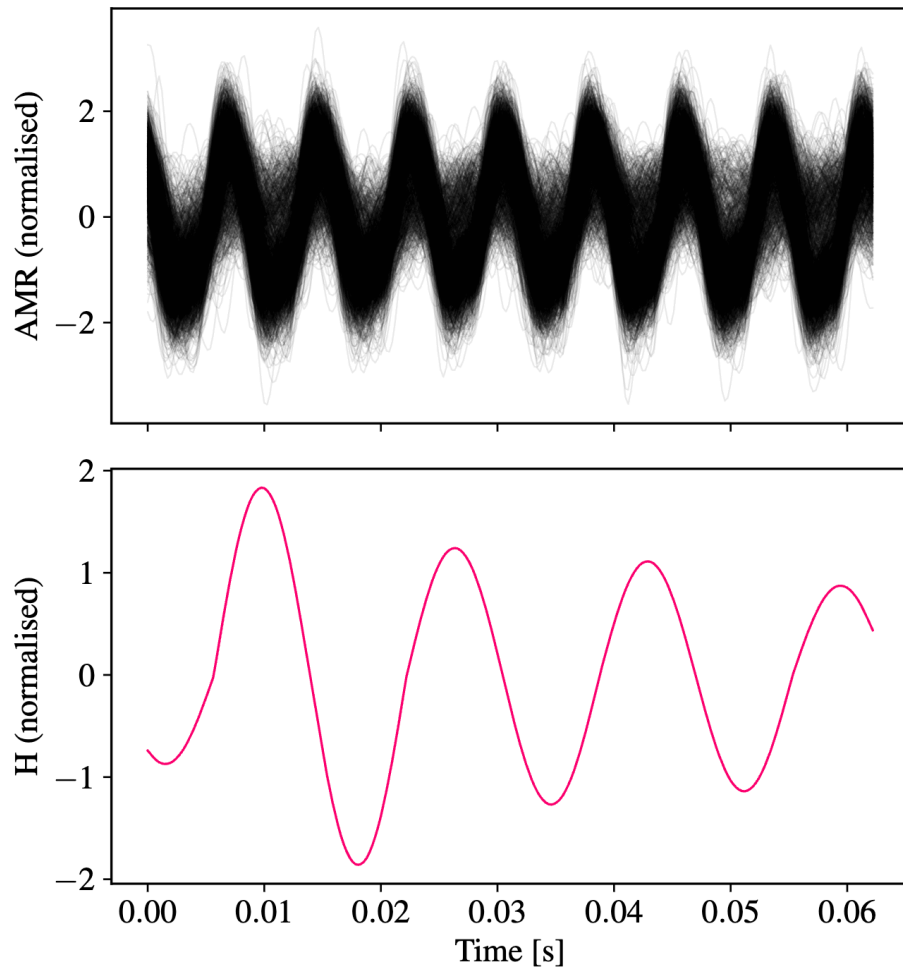


Fig. 1. Standardised AMR responses (top) from one applied H field (bottom). The data represents 5% of the overall trajectory.

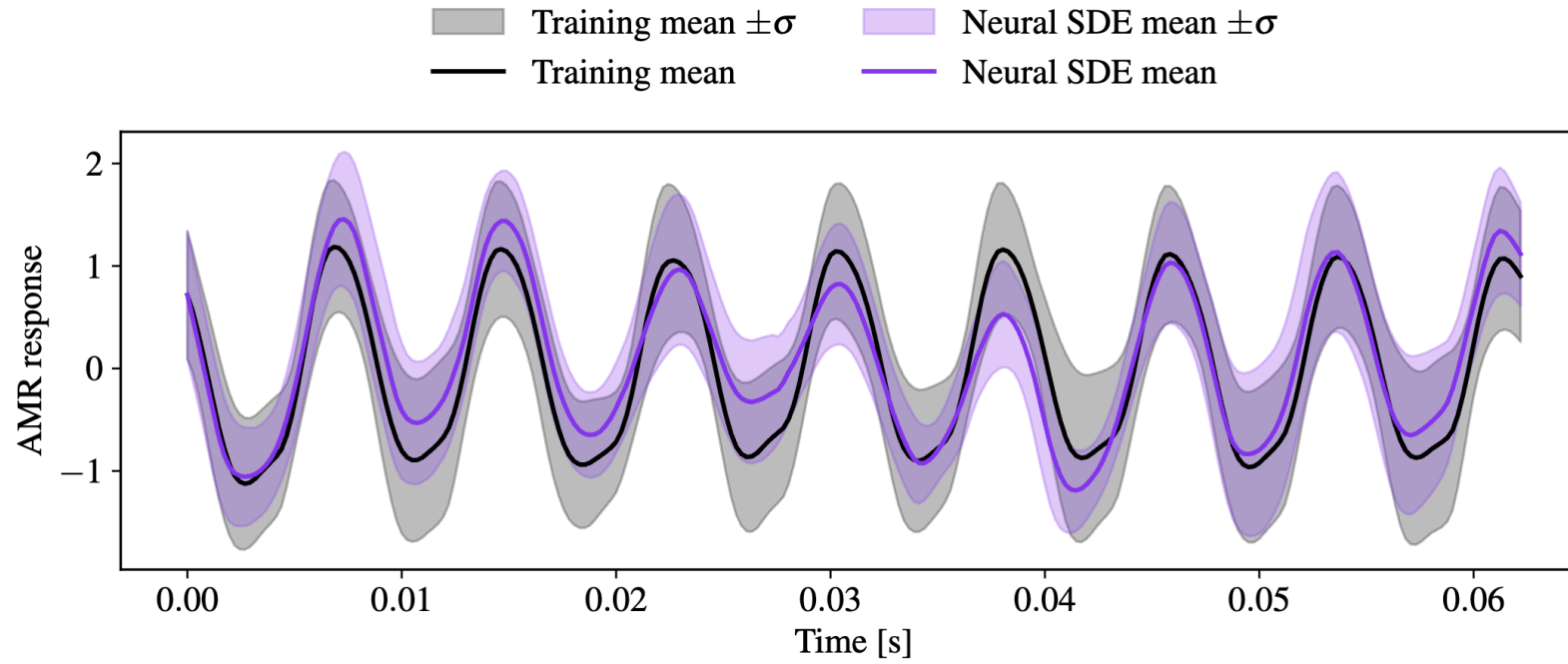
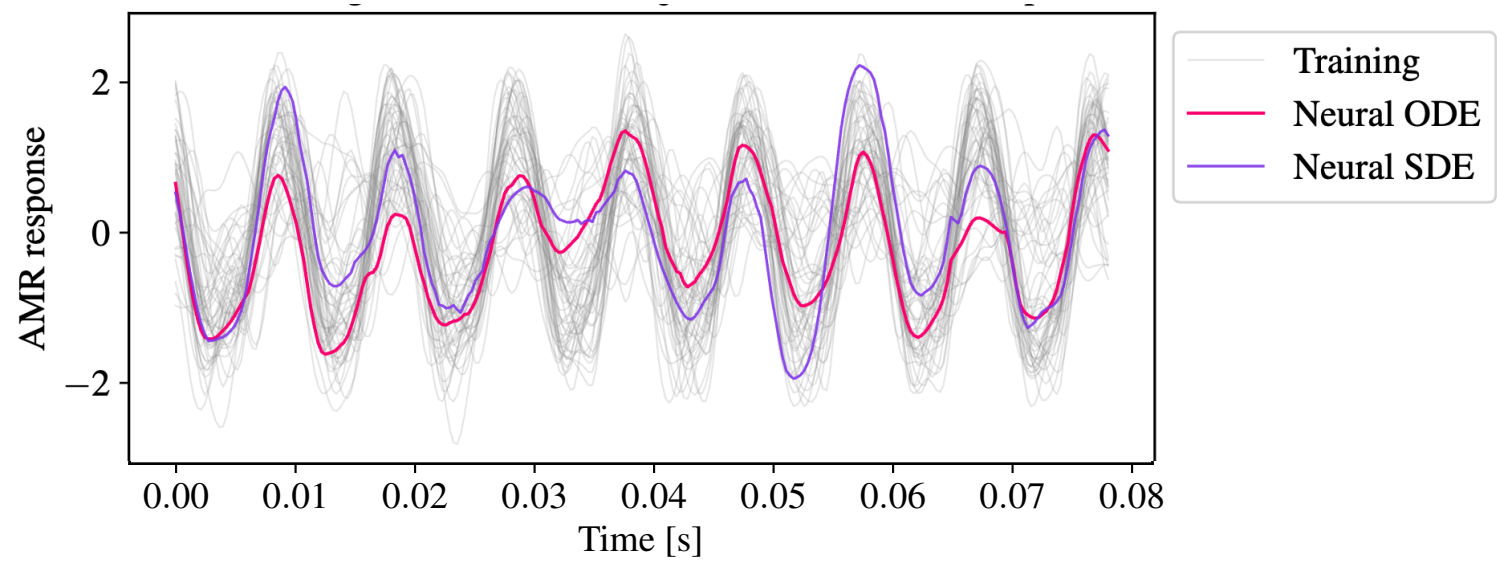


Fig. 2. Distributions of AMR responses used in the training data (mean in solid black, one standard deviation in shaded grey) and predicted AMR responses from the neural SDE model (mean in solid purple, one standard deviation in shaded purple). Close agreement is observed for a few periods. The distributions—whilst broadly following the same shape—begin to deviate further into the training. There are significant run-to-run stability issues—only once have we observed the neural SDE to follow broadly the correct distribution of trajectories.



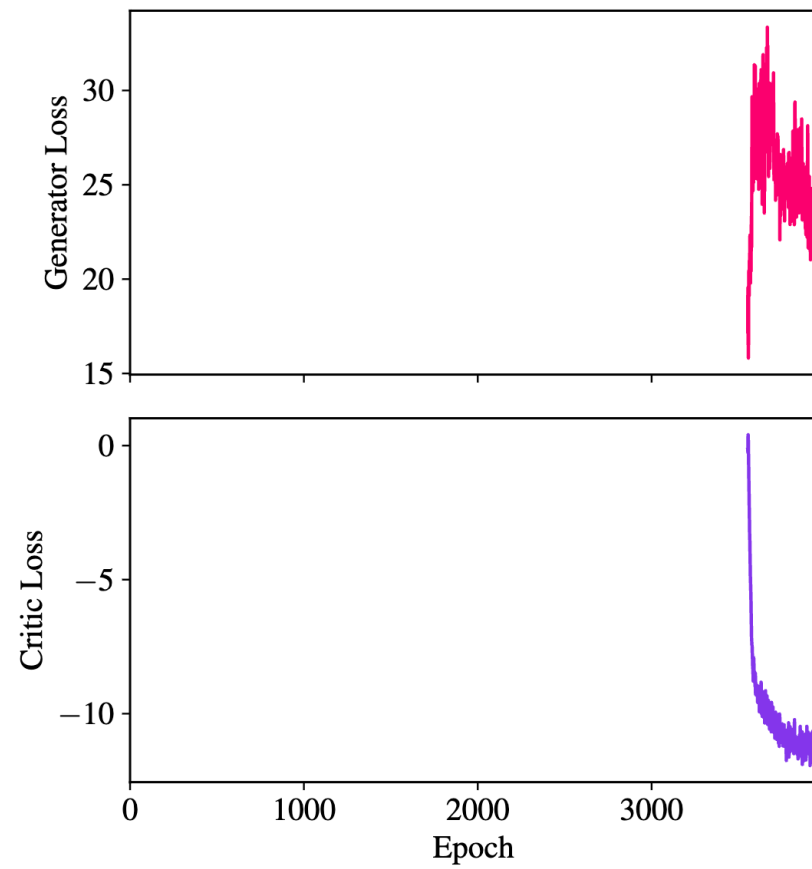


Fig. 3. SmoothL1 Loss (top [NEEDS FIXED]) for the neural ODE training. At epoch 4,086, the model parameters are frozen and are used as a baseline for the generative adversarial network training (middle and bottom).

Hyperparameters

Drift learning rate	3E-3
Diffusion learning rate	5E-4
Critic learning rate	5E-4
Generator learning rate	1E-4
Neural ODE epochs	4,086 (With early stopping patience = 150)
GAN epochs	400
Batch size	128 (Both Neural ODE and SDE)
Validation split	0.2
Patience (early stopping)	150
Critic window size	64
Critic updates per gen step	1
Gradient penalty weight	10
Moment matching weight	1.0
Drift net	(AMR, H, 2x harmonics, 5-step AMR context, 5-step H-field context) 3x hidden layers of 128 neurons Scalar output for dAMR/dt
Diffusion net	Same as Drift net, but outputs one noise channel
Critic net	MLP with hidden layers of size 64, 32, ending in a Wasserstein score
Physical units conversion:	$\text{AMR}_{\text{physical}} = \text{run_mean} + \text{z_norm} * \text{AMR_std}$